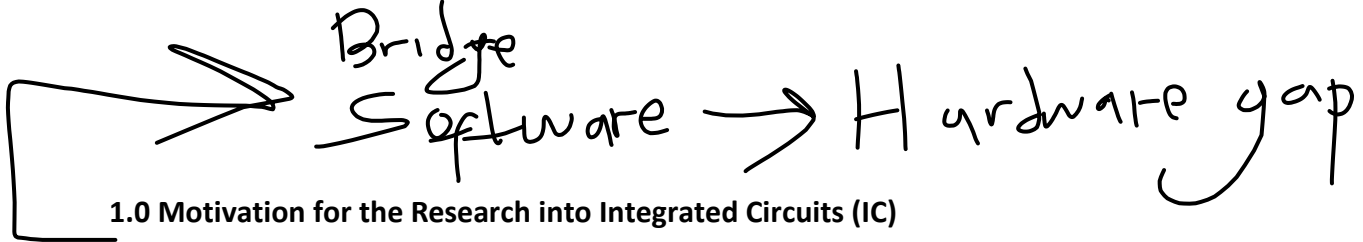


COMPUTER HARDWARE

LESSON ONE: INTEGRATED CIRCUIT (IC)

Video Presentation



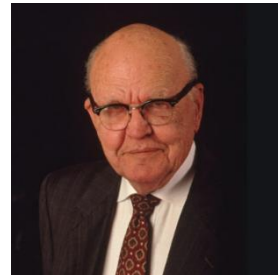
1.0 Motivation for the Research into Integrated Circuits (IC)

The motivation for research into ICs are:

- The gargantuan size of the first generation mainframe computers circuit that employed vacuum tubes, which consumes high electricity power.
- The software industries were developing programs at a very rapid growth rate with much sophistication; these gains will be stunted and of no use if there is no matching technological innovation in the electronics industries to manufacture computer hardware with the required processing power.

2.0 History of Integrated Circuits

- ✚ Invented jointly by Jack Kilby (1923-2005) & Robert Noyce (1927 – 1990)
- ✚ Both men worked independently and there were no collaboration. However, it was more or less exactly the same time.
 - Kilby was working for Texas Instruments (USA)
 - Noyce was working for Fairchild Semi-conductor
- ✚ Kilby filed patent first but Noyce's patent was granted first. It became a legal tussle
 - The companies both agreed to share the idea and a share-patent granted both companies in the year 1960.
- ✚ The Integrated Circuit is the most important far-reaching technology developed in the 20th century.
- ✚ Both Jack Kilby & Robert Noyce inducted into the inventor Hall of fame (1982 & 1983) respectively



Jack Kilby (1923-



Robert Noyce (1927-1990)

3.0 Introduction to Integrated Circuits (ICs)

- It is a neat way of cramming hundreds, thousands, millions, or even billions of electronic components onto tiny chips of silicon no bigger than a fingernail.
 - Other names for Integrated circuit are Chip, Microchip, Computer circuit, microprocessor chip, silicon chip
- Integrated circuit (IC) is an assembly of microscopic electronic components (transistors, diodes, capacitors, and resistors) and their interconnections fabricated as a single unit on a wafer of semiconducting material, especially silicon.
 - The main component is the transistors
 - Early ICs of the late 1950s consisted of about 10 components on a chip 0.12 in. (3 mm) square.
 - Very large-scale integration (VLSI) vastly increased circuit density, giving rise to the microprocessor.
 - The first commercially successful IC chip (Intel, 1974) had 4,800 transistors; Intel's Pentium (1993) had 3.2 million, and more than a billion are now achievable
- IC was first used by the United States Air force.



Figure 1.1: chips Dual inline & (SOIC)

3.1 Fitting IC to a Circuit board

Figure 1.2 shows a bread circuit board with an IC inserted.

- The red wire is connected to pin (Vcc)
- The black wire is connected to GND
- Pin 1 is identified by the notch portion

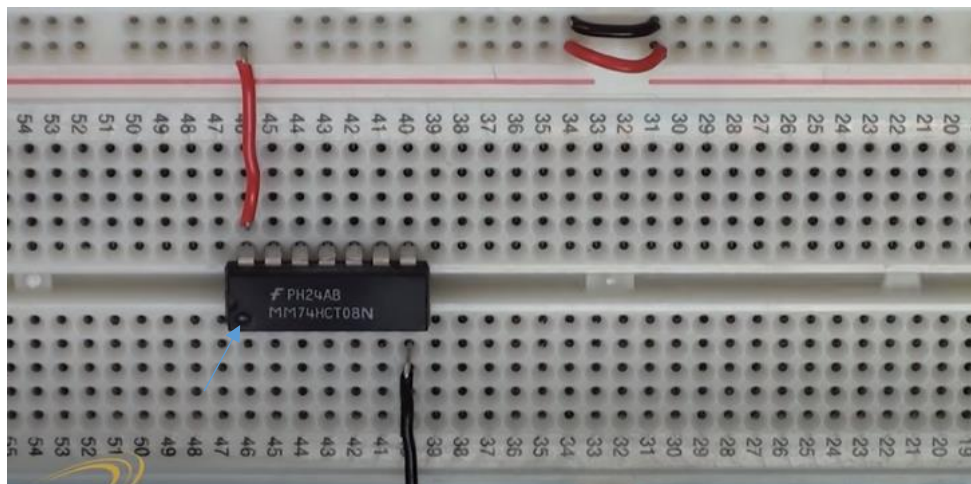


Figure 1.2: Breadboard showing how IC are fitted to Printed Circuit Board (PCB)

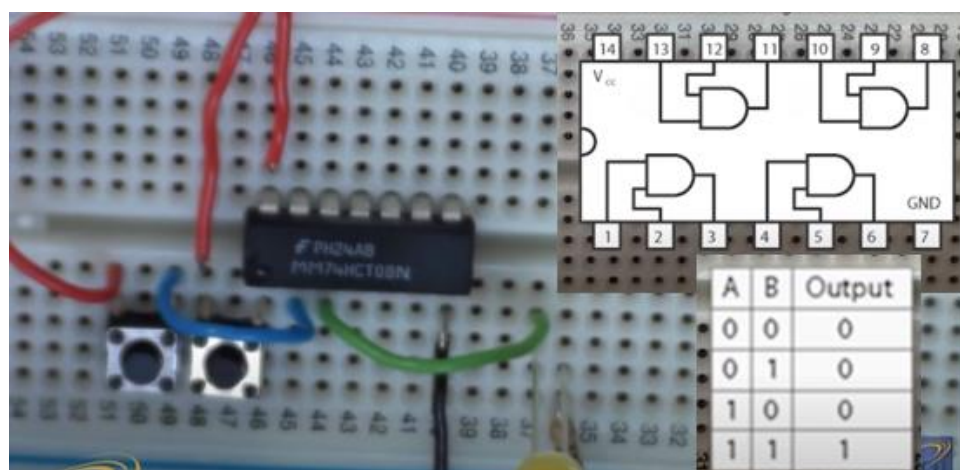


Figure 1.3: Breadboard showing more components connected

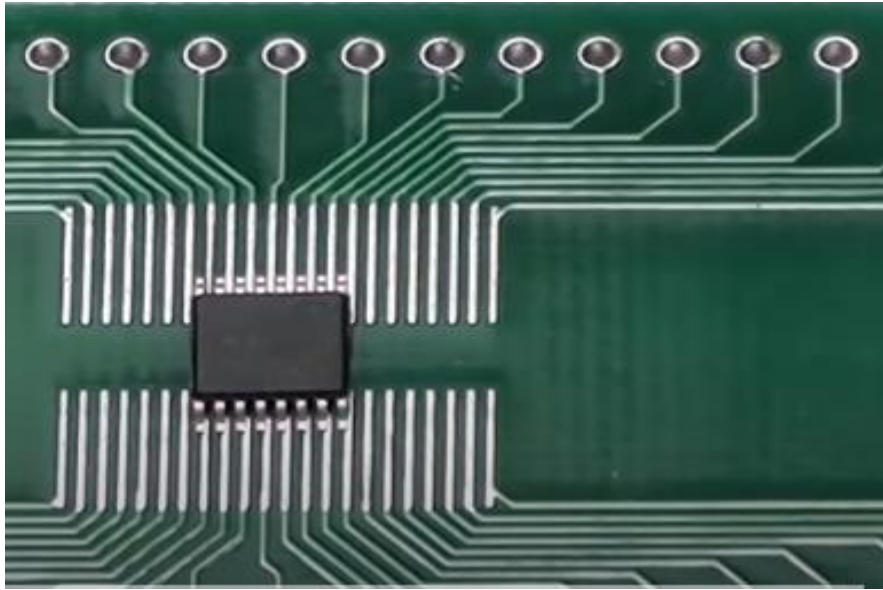


Figure 1.4: SOIC on a PCB

4.0 Types of IC

1. Analog IC

2. Digital IC

1. Analog IC is also called **linear IC**. Both input and output signal are continuous in nature. Output signal level is a function of the input signal level. Examples are: OP Amps, Voltage regulators and Comparators.

2. Digital IC operate on **digital data**. Logical 0 & 1; 0 = 0V and 1 = +5V. Examples are: Logic gates, counter, Flip-flops and (μ P).

5.0 Classification of Integration Circuit

Integration circuits are classified according to the following:

(1) The types of IC's employed and

(2i) The scale of Integration

1. Integrated circuits can be classified into two groups on the basis of the type of transistors, which they employ: bipolar integrated circuits, in which the principal element is the bipolar junction transistor;

Metal oxide semiconductor (MOS) integrated circuits, in which the principal element is the MOS transistor.

- Both depend upon the construction of a desired pattern of electrically active impurities within the semiconductor body, and upon the formation of an interconnection pattern of metal films on the surface of the semiconductor.
- Bipolar circuits are generally used where highest logic speed is desired, and MOS for largest-scale integration or lowest power dissipation. High-performance bipolar transistors and complementary MOS (CMOS) transistors have been

combined on the same chip (BiCMOS) to obtain circuits combining high speed and high density

2. Integrated circuits are also categorized according to the number of transistors or other active circuit devices they contain.
 - An IC is said to use small-scale integration (SSI) if it contains fewer than 10 transistors.
 - An IC that contains from 10 to 100 transistors is said to use medium-scale integration.
 - A large-scale integration (LSI) contains from 100 to 1,000 transistors
 - An IC that uses very-large-scale integration (VLSI) contains more than 1,000 transistors.

6.0 Usage of IC

Integrated Circuit can function as an Amplifier, Oscillator, Timer, Counter, Memory and Microprocessor (μP).

7.0 Discrete Vs Integrated Circuit

Discrete circuits are those circuits in which Bipolar Junction Transistor (BJT), Field Effect Transistor (FET) and diodes are interconnected along with resistors and capacitors to form electronic circuits.

8.0 What is Doping Semiconductors?

- Adding small quantities of impurities to a silicon atom is known as doping

Doping Silicon (14(2,8,4)) with Phosphorus (15(2,8,5)) = N-type (Electron predominates)

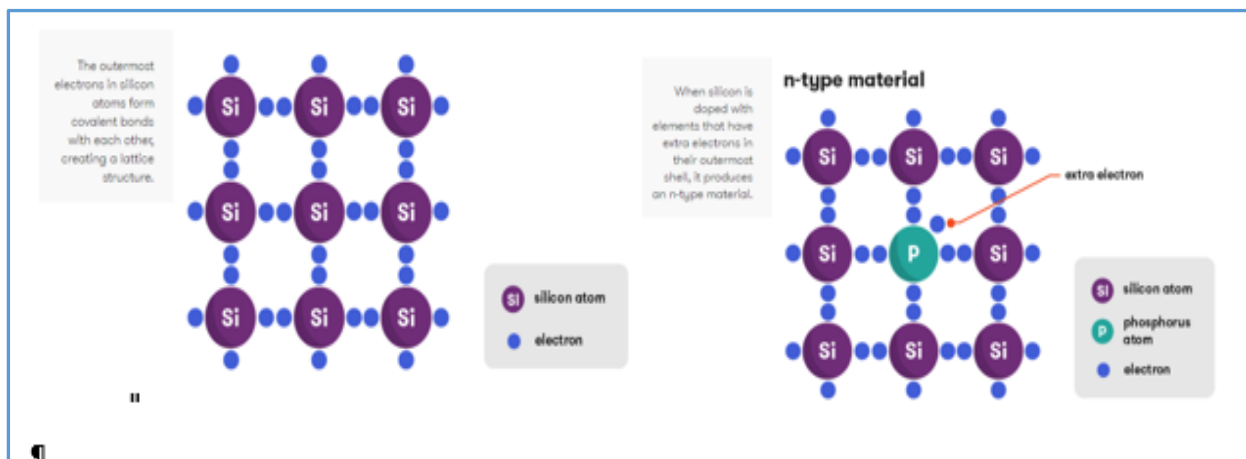


Figure1.5: Doping Silicon with Phosphorus – N-type

Doping Silicon (14 (2,8,4)) with Boron(5 (2,3)) = P-type (hole predominates)

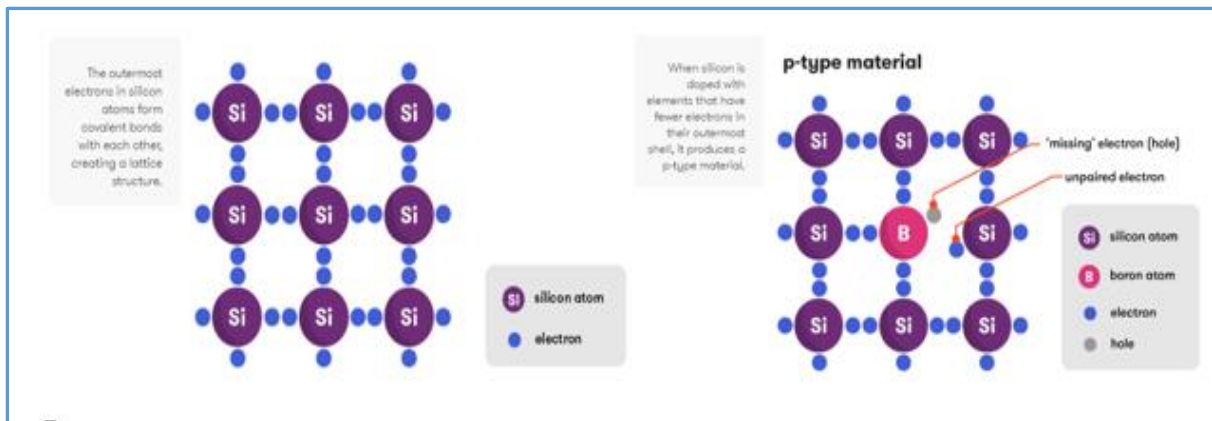


Figure 1.6: Doping Silicon with Boron – P-type

8.1 P–N Junction

A p–n junction is a boundary or interface between two types of semiconductor materials, p-type and n-type, inside a single crystal of semiconductor. The “p” side contains an excess of holes, while the “n” side contains an excess of electrons in the outer shells of the electrically neutral atoms there.

8.2 P-N Junction Diode

A PN-junction is formed when an N-type material is fused together with a P-type material creating a semiconductor diode. Figure 1.3 shows a P-N Junction diode.

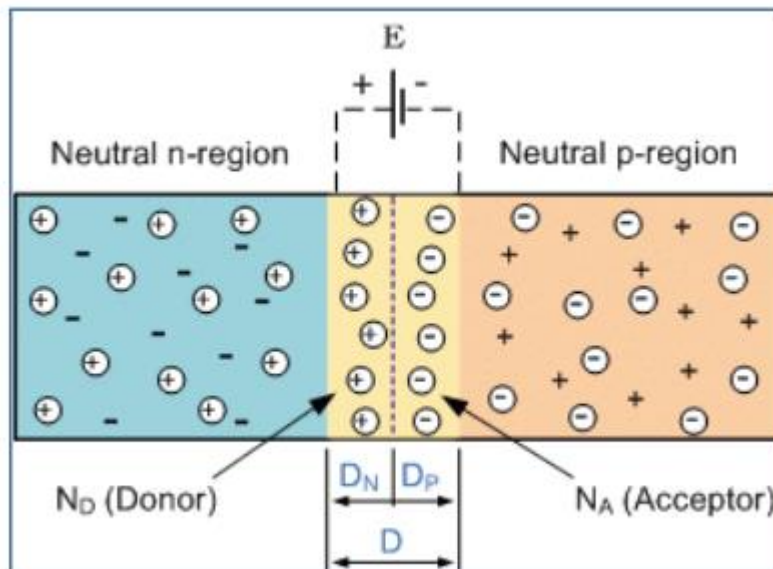


Figure 1.7: PN junction Diode

Question: What is the function of the diode in a circuit?

9.0 IC Fabrication Process

The process used to manufacture IC's is similar to that used to make components like a memory or processor chip for a computer. It all starts with a raw chemical element such as silicon, which is chemically treated or doped to make it have different electrical properties.

Doping is done by a variety of different processes:

- In one of them, known as sputtering, ions of the doping material are fired at the silicon wafer like bullets from a gun;
- Another process called vapor deposition involves introducing the doping material as a gas and letting it condense so the impurity atoms create a thin film on the surface of the silicon wafer;
- Molecular beam epitaxy is a much more precise form of deposition. Of course, making integrated circuits that pack hundreds, millions, or billions of components onto a fingernail-sized chip of silicon is all a bit more complex and involved more than it sounds.

Of course, making integrated circuits that pack hundreds, millions, or billions of components onto a fingernail-sized chip of silicon is all a bit more complex and involved more than it sounds.

A speck of dirt could cause serious havoc when working at the microscopic (or sometimes even the nanoscopic) scale in a laboratory. That is why semiconductors are made in spotless laboratory environments called clean rooms, where the air is meticulously filtered and workers have to pass in and out through airlocks wearing all kinds of protective clothing.

9.1 A summary of IC Fabrication Process

Greatly simplified, Figure 1.5 depicts the process involved in IC fabrication.

(a) The processes involved in IC fabrication

There are really only six separate steps (some of them are repeated more than once):

1. **Making wafers:** We grow pure silicon crystals into long cylinders and slice them (salami sliced) into thin wafers, each of which will ultimately be cut up into many chips.
2. **Masking:** We heat the wafers to coat them in silicon dioxide and use ultraviolet light (blue) to add a hard, protective layer called photoresist.
3. **Etching:** We use a chemical to remove some of the photoresist, making a kind of template pattern showing where we want areas of n-type and p-type silicon.
4. **Doping:** We heat the etched wafers with gases containing impurities to make the areas of n-type and p-type silicon. More masking and etching may follow.
5. **Testing:** Long metal connection leads run from a computer-controlled testing machine to the terminals on each chip. Any chips that do not work are marked and rejected.
6. **Packaging:** All the chips that work OK are cut out of the wafer and packaged into protective lumps of plastic, ready for use in computers and other electronic equipment.

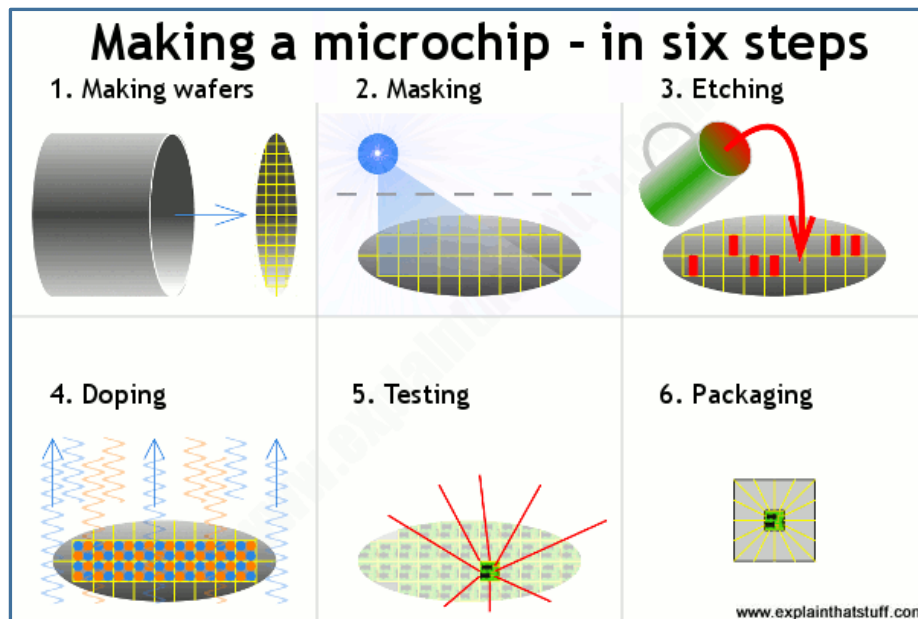


Figure 1.8: The process involved in IC fabrication

(b) Quality Control

Despite the controlled environment and use of precision tools, a high number of integrated circuit chips are rejected. Although the percentage of reject chips has steadily dropped over the years, the task of making an interwoven lattice of microscopic circuits and components is still difficult, and a certain amount of rejects are inevitable.

(c) Hazardous Materials and Recycling

The dopants gallium and arsenic, among others, are toxic substances and their storage, use, and disposal must be tightly controlled.

Because integrated circuit chips are so versatile, a significant recycling industry has sprung up. Many ICs and other electronic components are removed from otherwise obsolete equipment, tested, and resold for use in other devices.

End